

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location VA facility VA_way_1074595310 -- station KCXE, America/New_York
Committed pair OSM 1074588580 -> 1074588586
OSM way 1074588580 / OSM way 1074588586
Facade gap 103.5 m between the two halls
Weather record 43,539 real hourly records from KCXE
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3315 C at 120 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2024-04-14 (crossing)
Plant limit 18.0 C
Notice required 6 h
Forecast skill 0.00 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 12 of 24 hours with 2 mode changes.

Agent 12 of 24 hours free cooling, 2 mode change(s), 3 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 3 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	9.077	18.0	9.431	4.263
01	FREE	yes	-	9.467	18.0	8.651	3.516
02	FREE	yes	-	9.577	18.0	7.541	2.814
03	FREE	yes	-	9.577	18.0	6.541	2.489
04	FREE	yes	-	9.797	18.0	6.041	2.547
05	FREE	yes	-	9.857	18.0	5.431	2.422
06	FREE	yes	-	10.077	18.0	4.931	2.359
07	FREE	yes	-	10.687	18.0	9.371	2.458
08	FREE	yes	-	12.467	18.0	14.151	3.096
09	FREE	yes	-	14.333	18.0	18.610	3.897
10	FREE	yes	-	16.459	18.0	21.000	4.737

11	FREE	yes	-	17.759	18.0	22.780	5.205
12	mechanical	no	dry-bulb	18.607	18.0	24.441	5.559
13	mechanical	no	dry-bulb	22.942	18.0	25.567	5.359
14	mechanical	no	dry-bulb	24.977	18.0	26.500	4.348
15	mechanical	no	dry-bulb	27.416	18.0	27.500	3.724
16	mechanical	no	dry-bulb	26.792	18.0	27.220	3.150
17	mechanical	no	dry-bulb	26.472	18.0	26.791	3.292
18	mechanical	no	dry-bulb	26.094	18.0	25.890	3.591
19	mechanical	no	dry-bulb	24.993	18.0	23.933	3.572
20	mechanical	no	dry-bulb	25.017	18.0	22.761	4.057
21	mechanical	no	dry-bulb	25.463	18.0	22.111	4.348
22	mechanical	no	dry-bulb	25.547	18.0	21.110	4.606
23	mechanical	no	dry-bulb	25.766	18.0	20.000	4.610

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.077 C, which is 8.923 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.263 C, and the margin is measured, not chosen: 4.116 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.467 C, which is 8.533 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.516 C, and the margin is measured, not chosen: 3.370 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.577 C, which is 8.423 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.814 C, and the margin is measured, not chosen: 2.668 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.577 C, which is 8.423 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.489 C, and the margin is measured, not chosen: 2.342 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.797 C, which is 8.203 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.547 C, and the margin is measured, not chosen: 2.401 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.857 C, which is 8.143 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.422 C, and the margin is measured, not chosen: 2.276 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.077 C, which is 7.923 C

under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.359 C, and the margin is measured, not chosen: 2.213 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.687 C, which is 7.313 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.459 C, and the margin is measured, not chosen: 2.312 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.467 C, which is 5.533 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.096 C, and the margin is measured, not chosen: 2.950 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.333 C, which is 3.667 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.897 C, and the margin is measured, not chosen: 3.790 C of group-conditional forecast error for this hour of day, plus 0.1077 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by. *** THIS WAS WRONG: the intake actually reached 18.610 C, above the limit. The bound failed here, and it is counted as a breach rather than explained away. ***

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.459 C, which is 1.541 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.737 C, and the margin is measured, not chosen: 4.668 C of group-conditional forecast error for this hour of day, plus 0.0690 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by. *** THIS WAS WRONG: the intake actually reached 21.000 C, above the limit. The bound failed here, and it is counted as a breach rather than explained away. ***

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.759 C, which is 0.241 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.205 C, and the margin is measured, not chosen: 5.116 C of group-conditional forecast error for this hour of day, plus 0.0885 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by. *** THIS WAS WRONG: the intake actually reached 22.780 C, above the limit. The bound failed here, and it is counted as a breach rather than explained away. ***

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.607 C against a 18.0 C limit, so it fails by 0.607 C. It would take a limit 0.607 C higher, or a bound 0.607 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.941 C against a 18.0 C limit, so it fails by 4.941 C. It would take a limit 4.941 C higher, or a bound 4.941 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.977 C against a 18.0 C limit, so it fails by 6.977 C. It would take a limit 6.977 C higher, or a bound 6.977 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.416 C against a 18.0 C limit, so it fails by 9.416 C. It would take a limit 9.416 C higher, or a bound 9.416 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.792 C against a 18.0 C limit, so it fails by 8.792 C. It would take a limit 8.792 C higher, or a bound 8.792 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.472 C against a 18.0 C limit, so it fails by 8.472 C. It would take a limit 8.472 C higher, or a bound 8.472 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.094 C against a 18.0 C limit, so it fails by 8.094 C. It would take a limit 8.094 C higher, or a bound 8.094 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.993 C against a 18.0 C limit, so it fails by 6.993 C. It would take a limit 6.993 C higher, or a bound 6.993 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.017 C against a 18.0 C limit, so it fails by 7.017 C. It would take a limit 7.017 C higher, or a bound 7.017 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.463 C against a 18.0 C limit, so it fails by 7.463 C. It would take a limit 7.463 C higher, or a bound 7.463 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.547 C against a 18.0 C limit, so it fails by 7.547 C. It would take a limit 7.547 C higher, or a bound 7.547 C tighter, to change this hour.

23:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.766 C against a 18.0 C limit, so it fails by 7.766 C. It would take a limit 7.766 C higher, or a bound 7.766 C tighter, to change this hour.

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